

Ayan Nanda

Curriculum Vitae

ayan-cosmos.github.io/Personal-Web-Page

PERSONAL DETAILS

Birth	September 5, 1998
Address	Sauri,Paschim Medinipur,721466,West Bengal,India
Mobile	+91-9547760359
email	ayan.nanda@niser.ac.in, nandaayan97@gmail.com

EDUCATIONAL DETAILS

Master of Science (M.Sc.) in Physics 2020
Harish-Chandra Research Institute (HRI), Prayagraj, India
Affiliated to Homi Bhabha National Institute (HBNI)
First Class; Aggregate: **72.5%**

Bachelor of Science (B.Sc.) in Physics Honours 2018
Midnapore College (Autonomous), Midnapore, India
Affiliated to Vidyasagar University
First Class; Aggregate: **80.87%**

Higher Secondary Education 2015
Belda Gangadhar Academy, West Bengal, India
West Bengal Council of Higher Secondary Education
Percentage: **90.0%**

Secondary Education 2013
Sauri Bholanath Vidyamandir, West Bengal, India
West Bengal Board of Secondary Education
Percentage: **88.57%**

PHD DETAILS

PhD in Physics Dec,2020-Present
National Institute of Science Education and Research (NISER), Bhubaneswar, India
Supervisor: Dr. Nishikanta Khandai
Research Area: Large-Scale Structure, Galaxy Formation & Evolution.
Predoctoral Project: *Universality of Growth Function in an Expanding Universe*
PhD Thesis: *Evolution of Halo Shapes and the Role of Mergers and Environments*

MSC PROJECTS

1. Landau Levels in a Two-Dimensional Lattice System *Numerical Project* Supervisor: Dr. Pinaki Majumdar
2. Gravitational Waves *Master's Project* Supervisor: Dr. Anshuman Maharana

RESEARCH INTERESTS

Computational Cosmology; Large-Scale Structure; Dark Matter Haloes; Halo-Galaxy Connection; Galaxy Formation and Evolution; Cosmological Simulations; Predictions from Cosmological Simulations and their Comparison with Cosmological Observations.

PUBLICATIONS

First-Author Publications:

- **Ayan Nanda**, Nishikanta Khandai, Jasjeet Singh Bagla, Swati Gavas, **"Universality of Halo Shape and its Morphological Evolution across Cosmic Time"**, accepted in *JCAP* (2026). `arXiv:2603.26640`
- **Ayan Nanda**, Nishikanta Khandai, Swati Gavas, **"Multipole Characterization of Triaxial Dark Matter Haloes: A Spherical Harmonic Approach"**, manuscript in preparation.
- **Ayan Nanda**, Swati Gavas, Nishikanta Khandai, Jasjeet Singh Bagla, **"The Role of Mergers & Environments Affecting Halo Morphology and Morphology-dependent Clustering"**, manuscript in preparation.

Co-Author Publications:

- Abhinav Roy, **Ayan Nanda**, Nishikanta Khandai, **"Distribution of Gas in Main Sequence and Feedback Dominated Halos"**, manuscript in preparation.

- Swati Gavas, **Ayan Nanda**, Nishikanta Khandai, **"The effect of Finite Boxsize on Dark Matter Halo Shapes"**, manuscript under preparation.

SCHOOLS, WORKSHOPS & CONFERENCES

- **The Concurrence of Mega-Surveys**, ICTS, Bengaluru, India, 2026. Attended school and workshop. Delivered a project talk **Measuring Galaxy Shapes and Colours from Images**, under the supervision of Dr. Arun Kannawadi, followed by a talk in the next week based on my PhD work, titled **The Role of Mergers & Environments Affecting Halo Morphology and Morphology-dependent Clustering**.
- **ChaICA-6**, IISER Mohali, India, 2025. Delivered a contributed talk .
- **Indian Astronomical Society (ASI-2025)**, NIT Rourkela, India, 2025. Delivered a contributed talk **Self-Similarity of Halo Shapes in Cosmological Simulations**.
- **Summer School on Cosmology**, ICTP, Trieste, Italy, 2024. Presented a poster.
- **Largest Cosmological Surveys and Big Data Science**, ICTS, Bengaluru, India, 2023. Attended school and workshop.
  Delivered a project talk **Halo Splashback and Where to Find It** under the supervision of Dr. Susmita Adhikari.

ACADEMIC ACHIEVEMENTS & AWARDS

- Qualified **IIT-JAM 2018** with All India Rank (**AIR-53**).
- Qualified **JEST 2018** with All India Rank (**AIR-93**).
- Qualified **CSIR-UGC NET June,2019(LS)** with All India Rank (**AIR-39**).
- Qualified **CSIR-UGC NET December,2019(JRF)** with All India Rank (**AIR-100**).
- Qualified **GATE 2020** with All India Rank (**AIR-72**).
- Awarded **DST INSPIRE Scholarship**, Government of India (2015-2018).
- Awarded **DAE Fellowship**, Government of India (2018-2020).
- Awarded **Junior Research Fellowship**, NISER-DAE (2021-2023).
- Awarded **Senior Research Fellowship**, NISER-DAE (2023-Dec,2025).

TEACHING & MENTORING

- Teaching Assistant for **Astrophysics**.
- Teaching Assistant for **Electromagnetic Theory II**.
- Teaching Assistant for **Modern Physics Laboratory**.
- Mentored undergraduate and summer research students.

TECHNICAL SKILLS

Programming	FORTRAN, C, Python, Bash/Shell scripting
Cosmological Simulations	GADGET-4, AREPO, MP-Gadget, SUBFIND, ROCKSTAR, Consistent-Trees, SUBLINK
Simulation Experience	Experience in running and analysing large-scale cosmological N-body simulations
HPC	kalinga, astro, vikram (NISER)
Operating Systems	Linux
Optimised Computing	MPI, parallel computing, Slurm
Other	LaTeX, Git, GitHub

LANGUAGES

Bengali	Native
English	Professional proficiency
Hindi	Professional proficiency